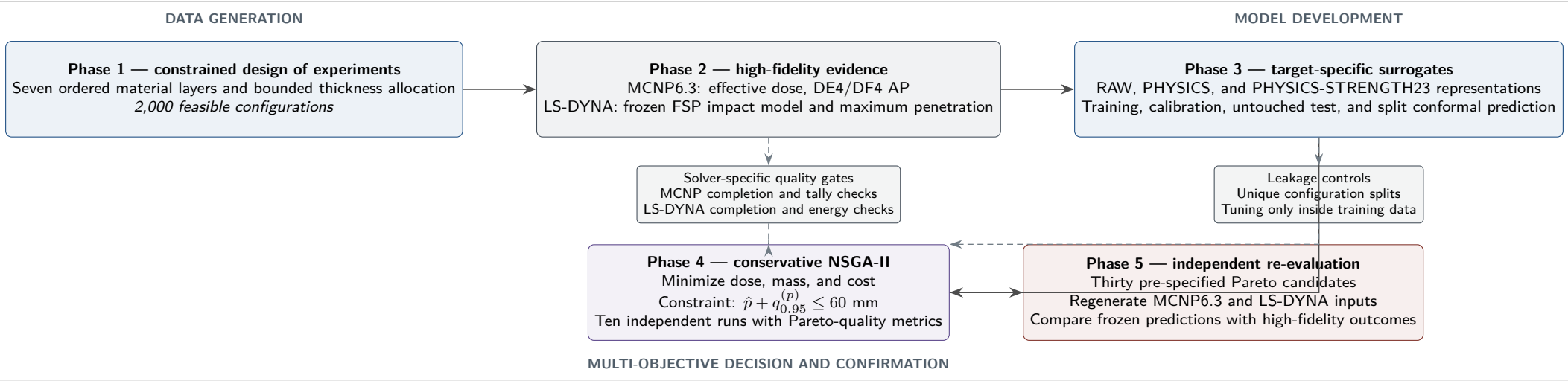

DATA GENERATION
MODEL DEVELOPMENT
Phase 1 — constrained design of experiments
Seven ordered material layers and bounded thickness allocation
2,000 feasible configurations
Phase 2 — high-fidelity evidence
MCNP6.3: effective dose, DE4/DF4 AP
LS-DYNA: frozen FSP impact model and maximum penetration
Phase 3 — target-specific surrogates
RAW, PHYSICS, and PHYSICS-STRENGTH23 representations
Training, calibration, untouched test, and split conformal prediction
Solver-specific quality gates
MCNP completion and tally checks
LS-DYNA completion and energy checks
Leakage controls
Unique configuration splits
Tuning only inside training data
Phase 4 — conservative NSGA-II
Minimize dose, mass, and cost
Constraint: $\hat{p} + q_{0.95}^{(p)} \leq 60$ mm
Ten independent runs with Pareto-quality metrics
Phase 5 — independent re-evaluation
Thirty pre-specified Pareto candidates
Regenerate MCNP6.3 and LS-DYNA inputs
Compare frozen predictions with high-fidelity outcomes
MULTI-OBJECTIVE DECISION AND CONFIRMATION